

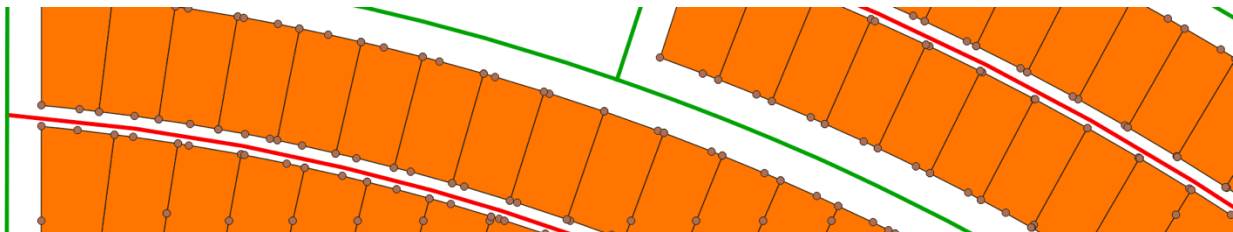
Question 1

Objective

To decide if we should only support lots with 4 straight sides.

Attempts

Before setbacks can be applied to define a buildable area, the front, side, and rear lot lines need to be identified. If we limit to lots with only 4 straight sides, this becomes a lot easier. The front is the closest to the street, the sides touch the front, and rear is remaining side. Only 19879 out of 29257 (67.9%) R1 parcels (R1 is for laneway homes) in Burnaby have 4 straight sides, however, there many lots that nearly qualify (refer to the image below).



If a lot has a side that isn't perfectly straight, it can be straightened with the Douglas-Peucker algorithm. This algorithm requires a tolerance parameter which can be thought of as the maximum allowable deviation between the initial line and the resulting line.

Evidence

Here is a summary of the number of 4-sided lots in Burnaby as tolerance changes.

Tolerance (m)	Number of 4-Sided Parcels	Percentage of R1 Parcels that are 4-Sided
None	19879	67.9
0.001	21839	74.6
0.01	22956	78.5
0.1	25140	85.9
1	26477	90.5

The complete rough notes can be found at the link below, but keep in mind these are just personal rough notes:

<https://docs.google.com/document/d/1cKiQvDWUyOiNcOgK01-6ccK5YBkRtHc4JmfTC8CLf-ss/edit?usp=sharing>

Diagnosis

This issue came up because it is difficult to define the front, sides, and rear lot lines in a robust way for many-sided lots.

Proposed Next Steps

Use a tolerance of 0.1 m so that limiting ourselves to 4-sided lots still captures 85.9% all R1 parcels. A larger tolerance could be used, but then we may want to add the tolerance value to any setbacks that are applied to ensure we error on the side of caution when calculating buildable area, this way any large deviations are accounted for. Lots with more than 4 sides can be flagged.

Decision Ask

Do you agree that we should limit ourselves to lots with 4 straight sides so that the front, side, and rear lot lines can be easily identified? If so, do you have thoughts on what tolerance (maximum allowable deviation) is acceptable to capture lots with 4 nearly-straight sides.

Question 2

Objective

To clarify how the rule extraction work and the source verification work should fit together before we expand the prototype.

One approach uses official bylaw text and an LLM to extract candidate rules. Another approach focuses on source matching, normalization, and verification before a rule is used in the final registry. These approaches are complementary, but we need to decide where the boundary should be.

The main question is whether LLM-extracted rules should be treated as final rules, or as candidate rules that still need source-backed verification.

Attempts

So far, we have explored two related directions:

1. LLM extraction pipeline: official bylaw source -> block cutting -> LLM extraction -> candidate rules -> second-round search for missed conditions/exceptions.
2. Source verification pipeline: researched workbook/source rule -> exact source snippet or table row -> compare value, condition, and operator -> normalized schema -> verification status -> deterministic checker.

Both directions seem useful, but they solve different parts of the project.

Evidence

The Surrey R1 extraction pipeline shows that LLMs can help generate candidate rules from official bylaw text. This is valuable because manual rule extraction is slow.

At the same time, the need for a second-round search to find missed conditions/exceptions suggests that first-pass LLM extraction can be incomplete. That is not a failure of the idea, but it means we need a verification step before using the extracted rules.

There are also normalization and generalization risks:

- Max lot coverage should usually be normalized as $\leq$, not equals, because it is a maximum limit.
- A rule value may be correct but attached to the wrong lot size, building type, unit type, or exception.
- A method that works on Surrey R1 may not automatically work on Burnaby or Vancouver because each city may use different bylaw structures and table formats.
- What works on a local Qwen/Ollama setup may not transfer exactly to GPT, Claude, Gemini, or another model. The rule schema, verification layer, and deterministic checker should be model-independent.

Diagnosis

The issue is not whether LLM extraction is useful. It is useful as a candidate-rule generator.

The risk is that a candidate rule can look correct while still being unsafe to use if it is missing a condition, exception, operator, or exact source reference. The calculation can be deterministic once the correct rule is selected, but rule selection itself is contextual.

For example, max lot coverage may depend on municipality, zone, lot size, building type, dwelling type, table row, and exceptions. If the system extracts only a percentage but not the context, it may apply the wrong rule to a parcel.

This is why source matching is important. RAG/source retrieval should not replace the rule registry or deterministic checker, but it can act as a support layer when direct extraction is uncertain, when sources disagree, when multiple chunks look similar, or when rules depend on exceptions and cross-references.

Proposed Next Steps

We suggest combining the two directions into one pipeline:

```
LLM extraction
-> candidate rules
-> second-pass search for missing context
-> source matching to exact snippet/table row
-> schema normalization
-> verification status
-> deterministic checker
```

For the next benchmark, we suggest testing one rule type first: max lot coverage across Surrey, Burnaby, and Vancouver.

For each city, we should check whether the pipeline correctly captures:

- source block or table row
- value
- condition
- operator
- building type
- exception if any
- verification status

This would help us evaluate whether the approach generalizes beyond one Surrey R1 case before we expand to more rule types.

Decision Ask

Should our final architecture treat LLM extraction as a candidate-rule generation step, followed by source-matching verification before rules are added to the final rule registry?

We also want to confirm whether max lot coverage across Surrey, Burnaby, and Vancouver is a good first benchmark before expanding to more rule types.